

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

M.Tech. (CSE) Sem - II Examination, 2011
EC – 708 Advanced Optical Communication

Date: 16.05.11, Monday

Time: 10.00 a.m to 1.00 p.m

Maximum Marks: 70

Instructions:

1. Attempt all questions.
2. Answer to the two sections must be written in the separate answer books.
3. Figures to the right indicate *full* marks.

SECTION- I

Q.1

- A A Double-heterojunction InGaAsP LED emitting at a peak wavelength of 1310nm has radiative and non radiative recombination time of 30 and 100ns, respectively. The drive current is 40mA. Find recombination lifetime and internal quantum efficiency. 3
- B A given silicon avalanche photodiode has a quantum efficiency of 65 percent at a wavelength of 900nm. Suppose 0.5 μ W of optical power produces a multiplied photocurrent of 10 μ A. find the multiplication factor M. 3
- C Define: Absorption, Spontaneous emission & Stimulated emission 3
- D Draw Photodiode pulse responses under various detector parameters 3

Q.2

- A Explain pin photodiode with schematic representation 6
- B Draw & Explain VAD method for fiber fabrication 5
- OR**
- A Explain Reach-through avalanche photodiode structure 6
- B Draw & Explain Double- Crucible method for fiber fabrication 5

Q.3

- A Compare single mode and multimode step-index & graded-index optical fibers 6
- B Explain Surface-emitting LED 6

OR

- A Explain elements of an optical fiber transmission link 6
- B Explain Fabry-Perot cavity for a laser diode 6

SECTION – II

Q4

- A Which three Topologies are used for fiber optic network? 3
- B What is atmospheric attenuation? 3
- C Explain public network application of optical fiber 5

Q.5

- A Explain SONET/SDH Rings 7
- B Explain Industrial application of optical fiber 5

OR

- A Explain SONET/SDH Networks 7
- B Explain military application of optical fiber 5

Q.6

- A Explain performance of WDM+EDFA Systems 6

OR

- A Explain idealized arrangement for unguided optical communication system 6
- B Explain any one example of unguided optical communication system 6